آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-II Chemistry Annual Examinations 2025

Introduction

This document has been produced for the teachers and candidates of the Higher Secondary School Certificate (HSSC) Part II Chemistry. It contains comments on candidates' responses to the 2025 HSSC-II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific examples of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs, which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used. It is imperative to refer to the command word guide available on the AKU-EB website for understanding the expectations of the command word.

General Observations

Overall, a significant number of candidates performed well in constructing responses that reflected their understanding of the subject and their ability to apply higher-order thinking skills to a variety of question types. Notable strengths included the classification of carbohydrates, the identification of the significance of spectroscopy—both in general and across various fields—and the explanation of sewage water treatment stages.

However, certain areas did not meet expectations. For instance, questions related to the IUPAC and trivial names of phenols, the identification of chiral centres, and the preparation of Grignard's reagent were often answered inadequately. The latter particularly required a detailed explanation of the preparatory procedure along with the rationale for the specific conditions used as precautionary measures.

To support candidates in strengthening their conceptual understanding, teachers should focus on the following areas:

- Mastery of IUPAC nomenclature rules, including the identification of specific and multi-functional groups during naming.
- Clear differentiation between cracking and fractional distillation processes.
- Accurate identification of chiral carbons/ centres.

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS

Constructed Response Questions (CRQs)

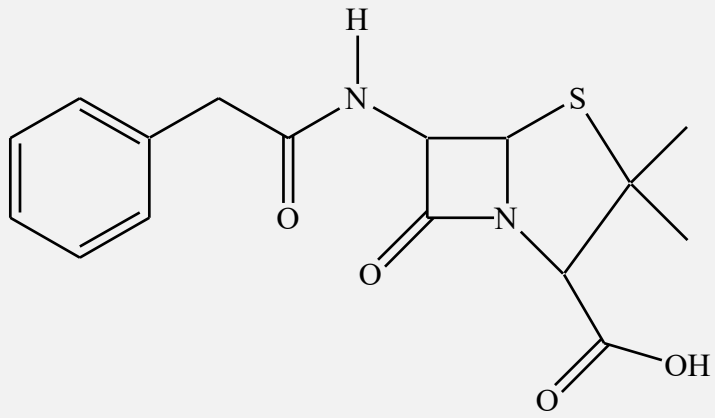
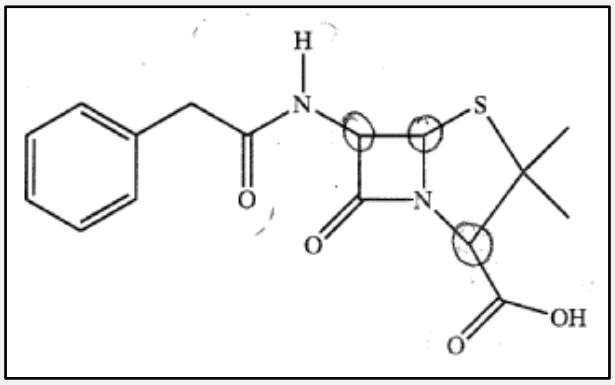
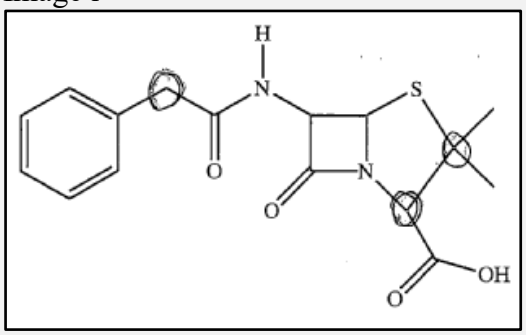
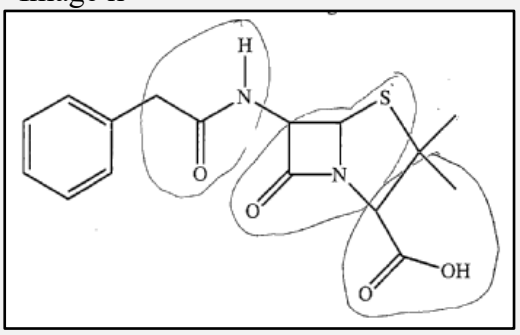
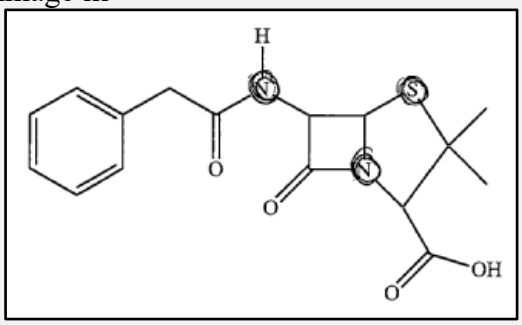
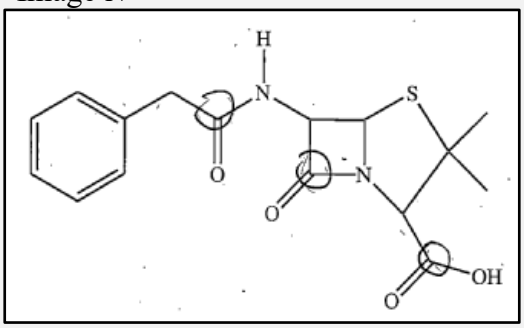
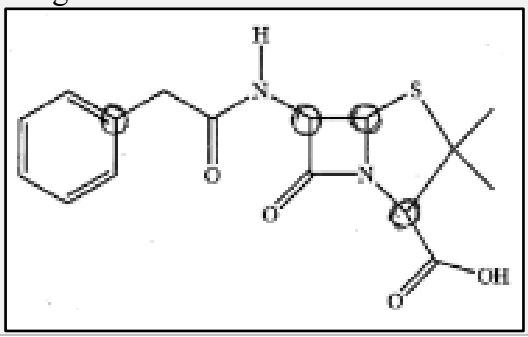
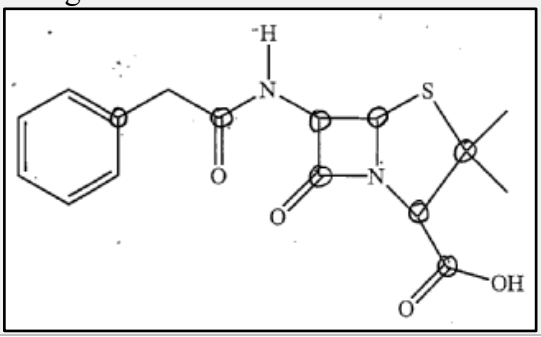
Question No. 1	
Question Text	Encircle THREE chiral centres in the given structure of an organic molecule. 
SLO No.	15.3.4
SLO Text	Determine chiral centres in the structural formula of a molecule.
Max Marks	3
Cognitive Level	A*
Checking Hints	1 mark for each correct determination of chiral centre (3 required) If the candidate encircles; • any 3 carbons, 1 mark each for the correct circle • any 4-6 carbons, 2 marks for 3 correct, 1 mark for 2 correct and zero marks for 1 correct circle • any 7-8 carbons, 1 mark for 3 correct, 0 mark for 2 or 1 correct circle • more than 8 carbons, hence no clarity of concept, so zero marks
Overall Performance	Overall, most candidates answered the question satisfactorily; however, few demonstrated a clear understanding of the differences between chiral carbons, carbonyl carbons, heteroatoms, and functional groups in a macromolecular structure. The question proved to be diagnostic, revealing a broad range of misconceptions in the identification and differentiation of fundamental structural features, along with over-citation of the keyword.
Description of Better Responses	*Better responses* demonstrated accurate identification of the chiral centres within the given molecule. In several cases, candidates went a step further by illustrating the fourth bond of the carbon, often adding a hydrogen atom, to confirm its tetrahedral nature and establish its chirality. These candidates also paid close attention to the instruction in the question, correctly noting that the word 'THREE' was explicitly emphasised, and thus appropriately encircled only three carbon atoms in the structure.

Image of Better Response	
Description of Weaker Responses	A significant number of *weaker responses* were observed for what was a relatively straightforward question. Some candidates attempted to identify three chiral centres but included only one or two correct ones. A common error was confusing chiral centres with carbonyl carbons; others mistakenly identified heteroatoms or functional groups, likely because the structure contained three of each. A more concerning trend was the disregard for the instruction to identify *only* three atoms, despite 'THREE' being clearly emphasised. Several candidates encircled four to eight atoms or more, seemingly expecting examiners to determine which were correct. This guessing approach undermined the question's intent and resulted in a mark loss. For example, encircling four atoms reduced the likelihood of full accuracy to 25%, making it improbable to earn even one mark. With six selections, this dropped to 50%, and with more than six, it indicated a complete lack of conceptual clarity, resulting in reduced or no credit at all.
Images of Weaker Response	Image i  Image ii  Image iii  Image iv  Image v  Image vi 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy** Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

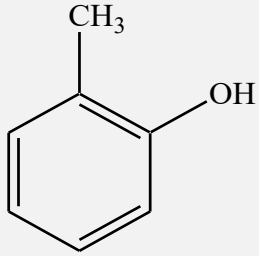
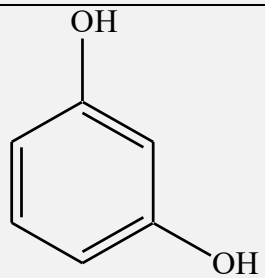
Any Additional Suggestion: Teachers should provide students with a range of organic structures to practise identifying chiral centres. The use of 3D models can help visualise the presence of implicit hydrogen atoms not shown in skeletal structures. Additionally, coloured playdough can be used to represent four distinct groups around a chiral carbon, while matchstick models may aid in reinforcing spatial orientation and conceptual understanding.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

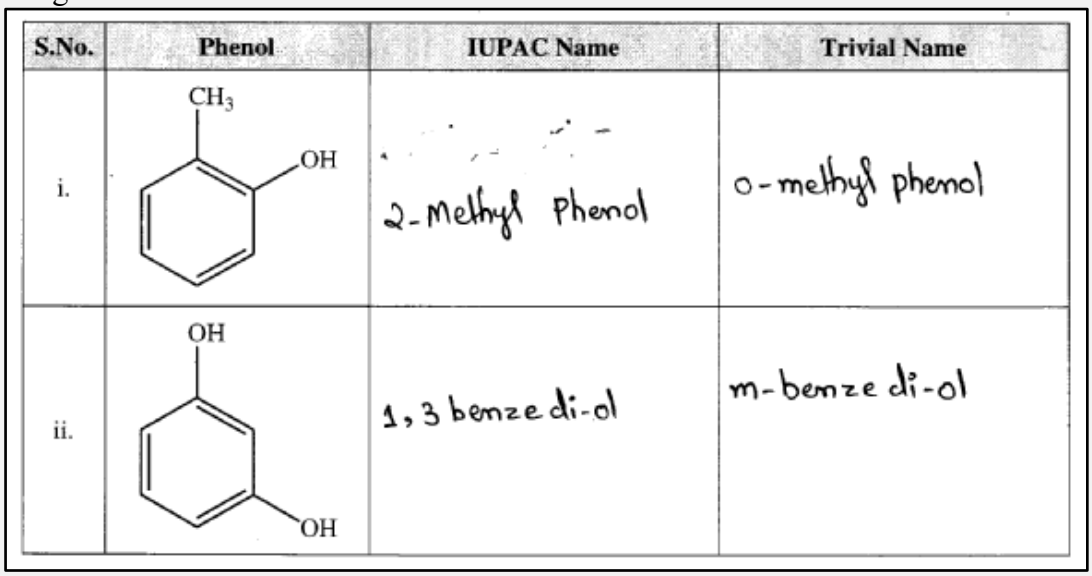
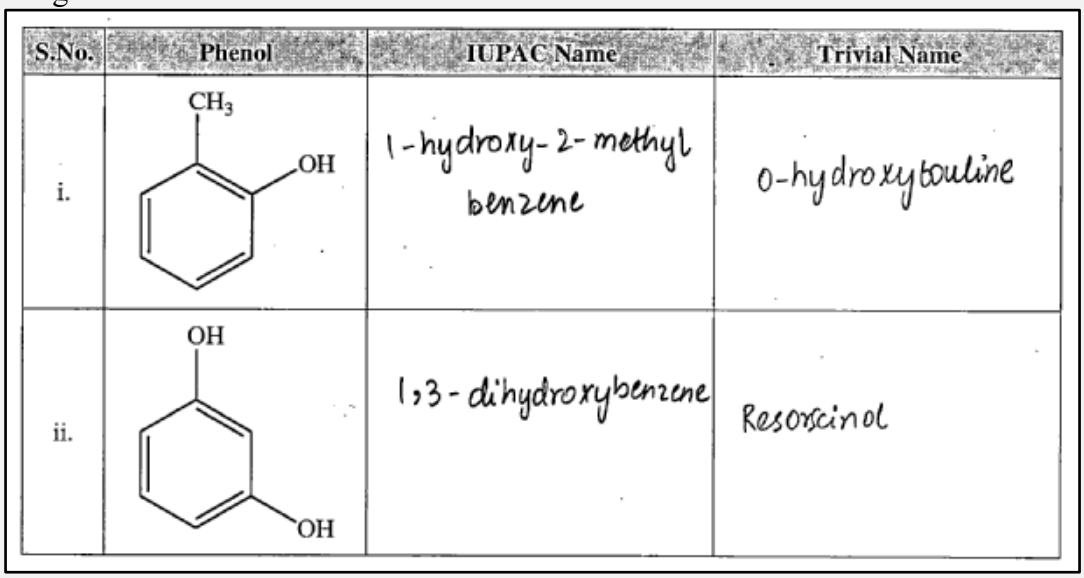
Question No. 2

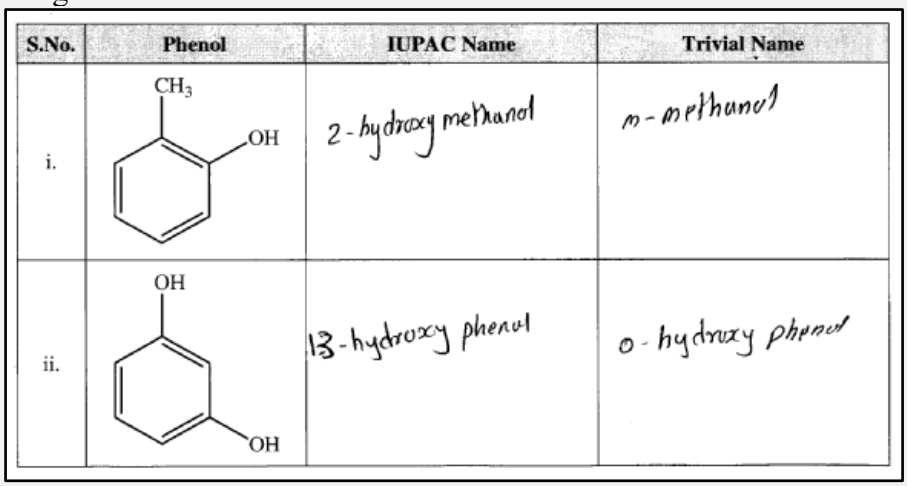
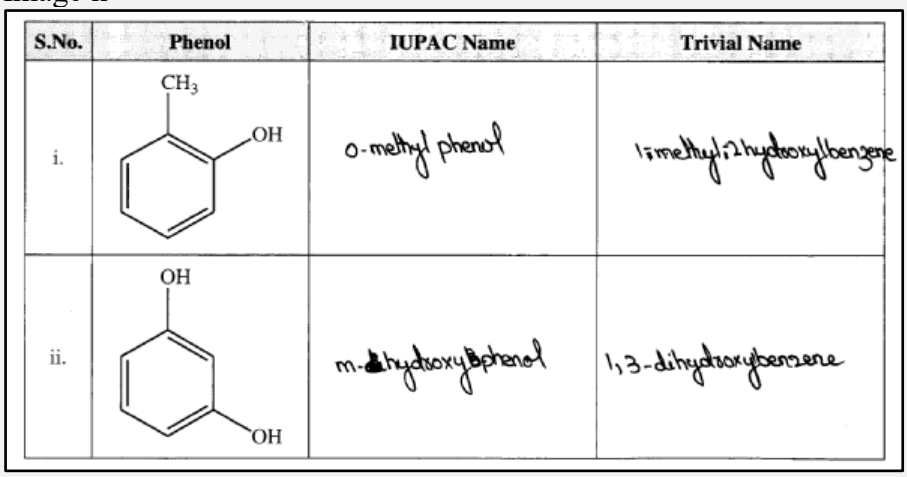
Question Text

Complete the given table about the nomenclature of phenols.
(**Note**: IUPAC-International Union of Pure and Applied Chemistry)

S.No.	Phenol	IUPAC Name	Trivial Name
i.			
ii.			

SLO No.	18.2.1
SLO Text	Apply IUPAC and trivial system for naming different phenols.
Max Marks	4

Cognitive Level	A
Checking Hints	1 mark for each name (4 required)
Overall Performance	Overall, the candidates' performance in this question concerning the simplest IUPAC nomenclature at the HSSC II level remained unsatisfactory.
Description of Better Responses	*Better responses* demonstrated a clear understanding of the rules for naming phenols using both IUPAC and trivial nomenclature. Most candidates accurately applied IUPAC conventions, including correct use of position numbers, hyphens between numbers and letters, commas between multiple numbers, and spacing between names of distinct functional groups. Examples included 1-hydroxy-2-methylbenzene and 1,3-dihydroxybenzene for the given structures. The corresponding trivial names—cresol and resorcinol—were also commonly provided. Additionally, many candidates correctly used the prefixes ortho- (o-) and meta- (m-) to indicate substituent positions on the benzene ring.
Images of Better Responses	Image i  Image ii 
Description of Weaker Responses	In *weaker responses*, candidates showed difficulty with IUPAC naming conventions. Many incorrectly used the common base word 'phenol' instead of applying systematic IUPAC rules. A frequent error was the interchange of naming patterns—using numerical positions in trivial names and prefixes such as *o-* and *m-* in IUPAC names. Several candidates also

	confused the two systems, often providing the common name in place of the IUPAC name and vice versa, particularly for Structure ii.
Images of Weaker Responses	Image i  Image ii 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy Used for this SLO**	**Assessment Strategies**
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Any Additional Suggestion: Teachers should begin with the basics of functional groups and hydrocarbon structures, using analogies and mnemonics to simplify concepts. Engage students with interactive activities like molecular modelling and peer collaboration to reinforce learning through regular quizzes and practice worksheets.

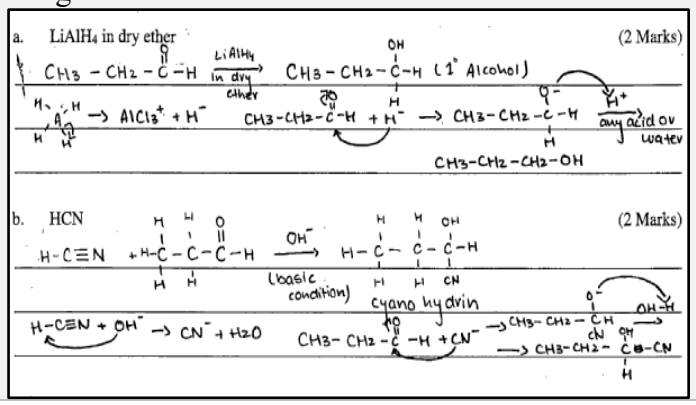
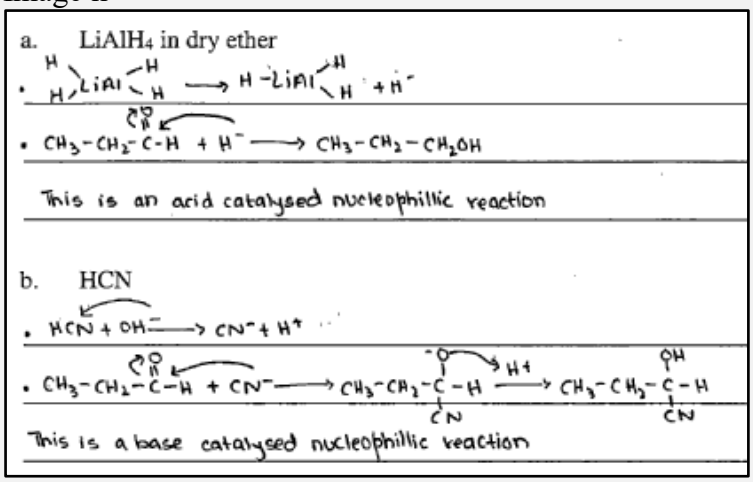
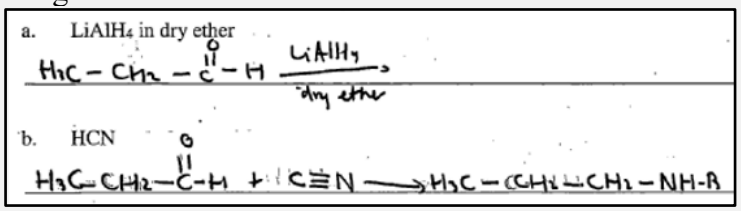
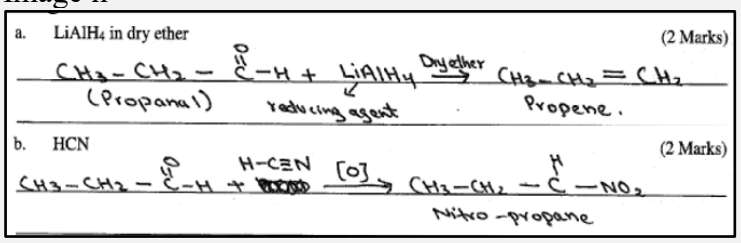
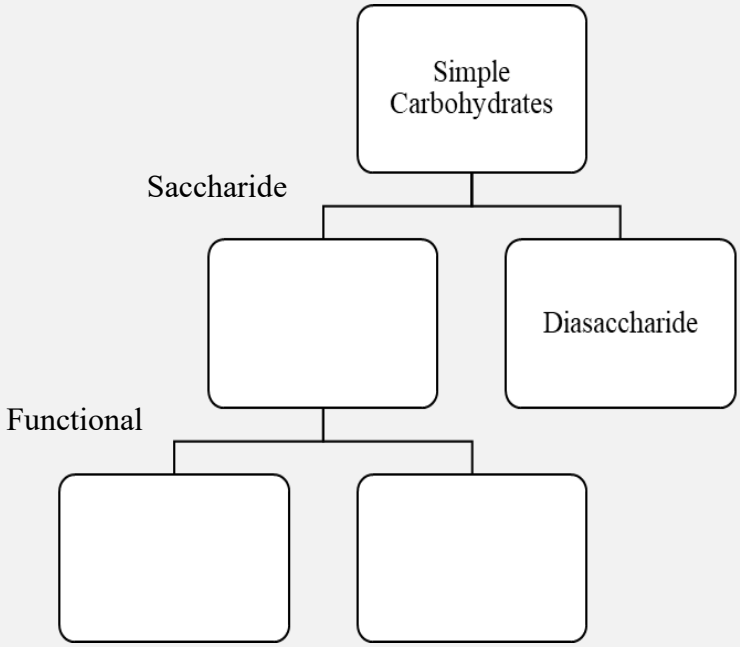
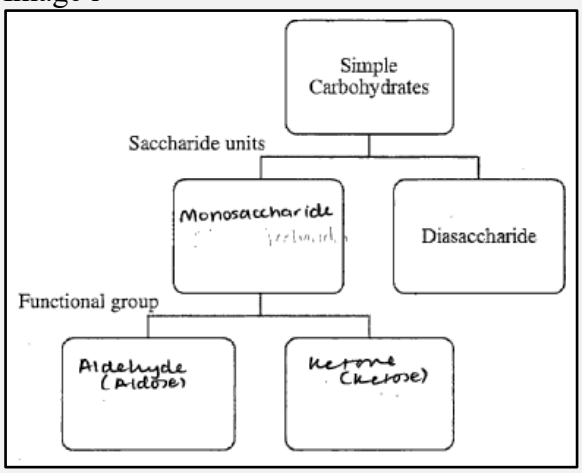
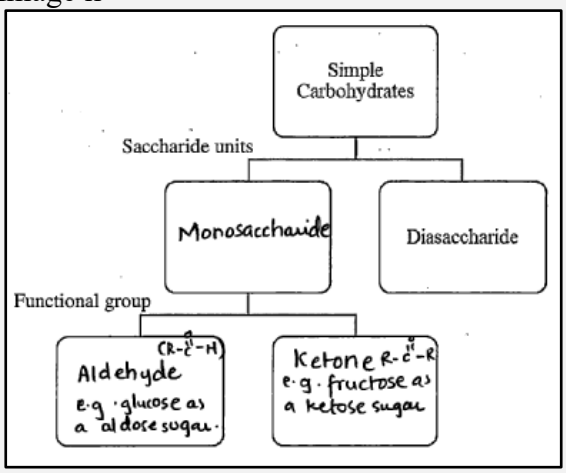
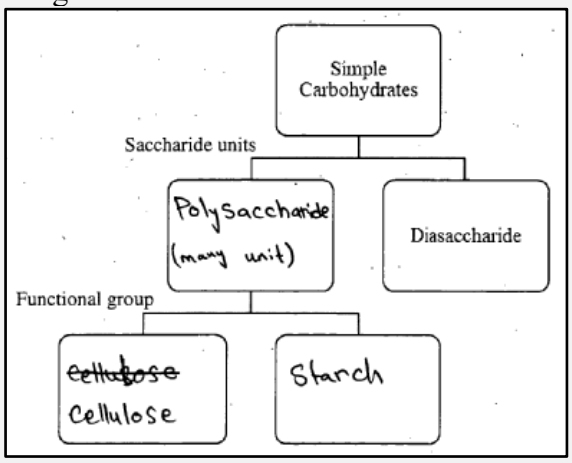
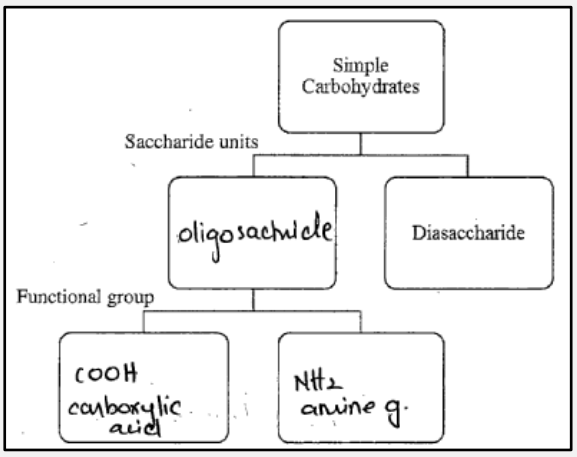
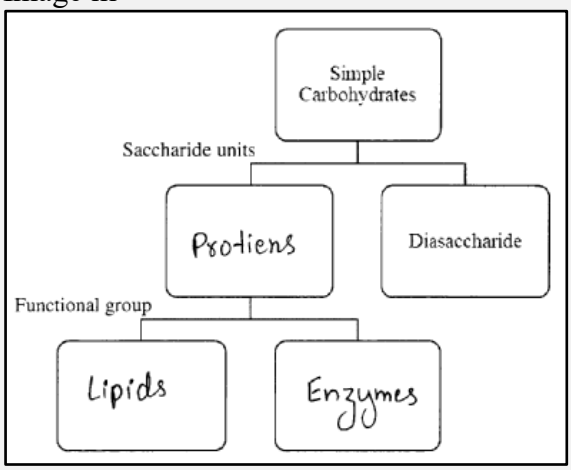
Question No. 3	
Question Text	Write a chemical equation to show the reaction of propanal with the following reagents. (**Note**: Mention the necessary reagents and conditions for a complete reaction.) a. $LiAlH_4$ in dry ether b. HCN
SLO No.	19.4.1, 19.4.3
SLO Text	Describe the base catalysed nucleophilic addition reaction of aldehydes and ketones, i.e. a. addition of hydrogen cyanide b. addition of Grignard reagent c. addition of sodium bisulphate d. Aldol condensation e. Cannizzaro's reaction f. haloform (iodoform) reaction; Describe the reduction of aldehydes and ketones using: a. Clemensen reduction method b. Wolff-Kishner reduction method c. hydride reagents d. carbon nucleophiles e. nitrogen nucleophiles f. oxygen nucleophiles;
Max Marks	4
Cognitive Level	U
Checking Hints	2 marks for each complete equation (2 required) Equation 1: 1 mark product + 1 mark reagents and given conditions including H_2O OR 1 mark product + 1 mark any one condition if the correct reactant is mentioned Equation 2: 1 mark product + 1 mark reagents and given conditions OR 1 mark product + 1 mark condition with OH^-/ base catalysed is mention
Overall Performance	The overall performance in this question was unsatisfactory. Instead of writing hydrolysis as the condition in an equation in part (a), candidates mentioned what is already in the question, i.e., dry ether.
Description of Better Responses	*Better responses* reflected a sound understanding of chemical reactions and equation writing. Many candidates correctly identified propanal as the common starting material in both equations. In addition to the provided conditions, several candidates accurately stated additional reaction conditions. For instance, in reaction (a), some included H_2O either as a condition on the arrow or within a mechanism, though not required, which indicated strong conceptual understanding. Similarly, in reaction (b), many candidates correctly identified NaCN/HCl as a reaction condition, while an equal number outlined the base-catalysed mechanism leading to the correct products in both cases.
Images of Better Response	Image i 

	Image ii 
Description of Weaker Responses	In *weaker responses*, most candidates were able to draw the correct structure of propanal in both reactions. A majority also omitted H_2O in chemical reaction (a), despite its relevance. Additionally, some candidates struggled to identify the reaction conditions in equation (b); however, they were still able to recall that the product was a cyanohydrin or otherwise.
Images of Weaker Response	Image i  Image ii 

Suggestions for improvement (Highlight all that apply)

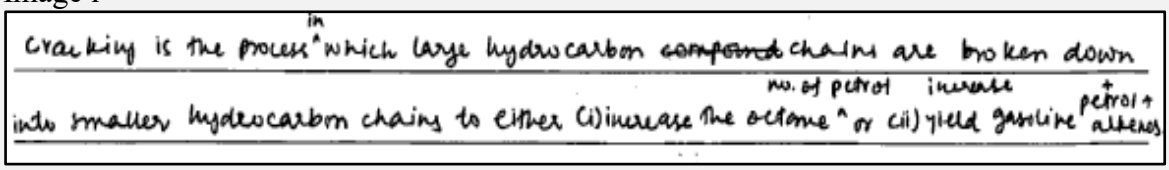
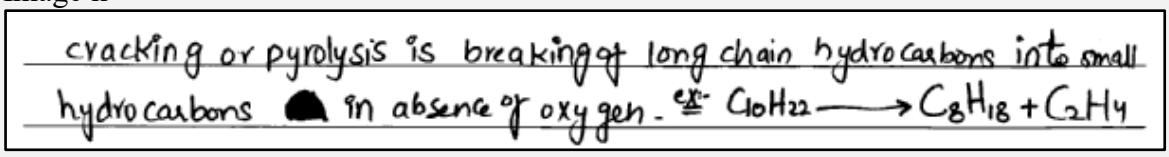
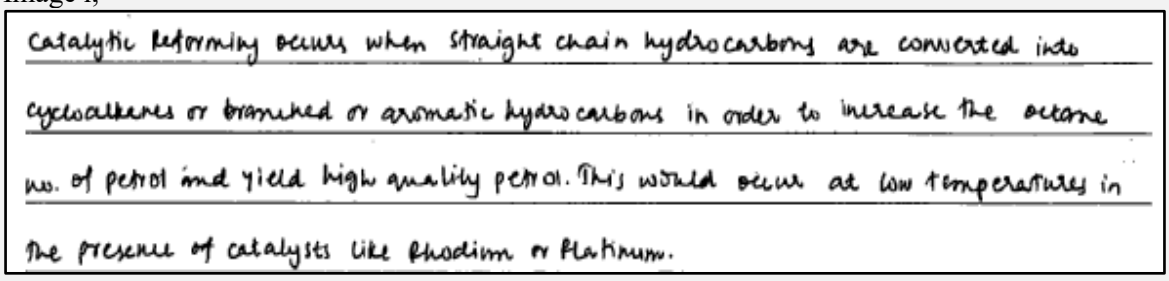
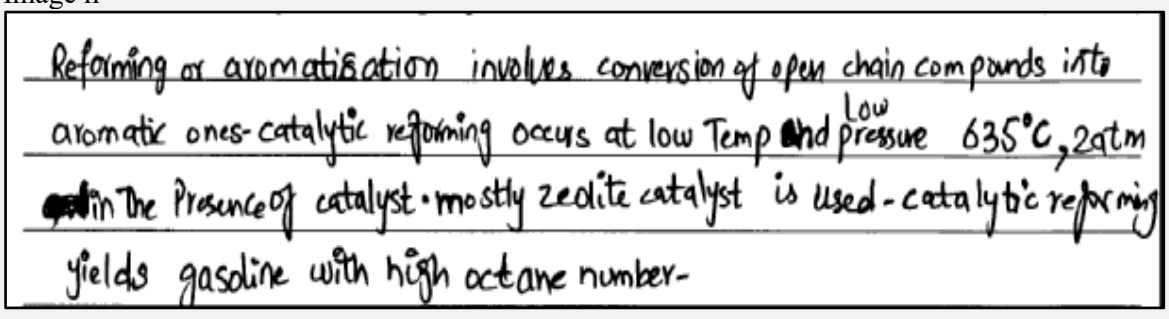
Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
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Any Additional Suggestion:		

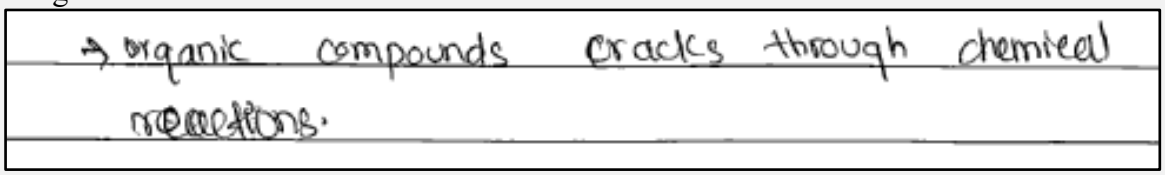
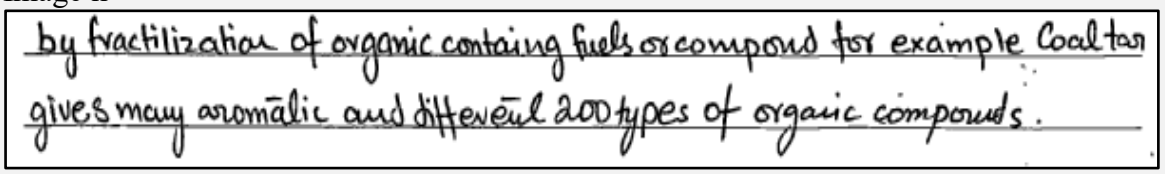
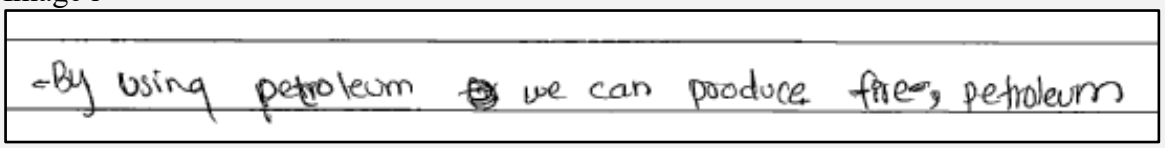
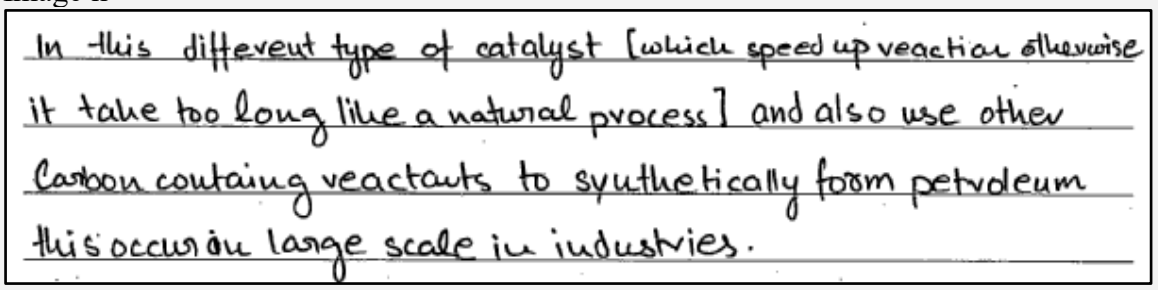
Question No. 4	
Question Text	Complete the given classification chart based on the given attributes. (**Note**: The first classification has been done for your convenience.) 
SLO No.	21.1.1
SLO Text	Describe the basis of classification of carbohydrates, proteins and lipids.
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for each correct classification (3 required)
Overall Performance	Overall, candidates performed satisfactorily on this question. Only a few were unable to classify carbohydrates based on the saccharide unit.
Description of Better Responses	*Better responses* accurately classified carbohydrates using two distinct criteria: saccharide units and functional groups. These candidates demonstrated a clear understanding that functional group-based classification applies only to monosaccharides. They followed the logical flow of the diagrammatic representation, correctly identifying monosaccharides in the missing box and further dividing it into aldoses and ketoses. While some responses mentioned the general formula of functional groups directly, which is acceptable but less desirable, others took it a step further by connecting the question to SLO 19.1.3, citing glucose and fructose as examples of aldehydic and ketonic carbohydrates, respectively.
Images of Better Response	Image i  Image ii 

Description of Weaker Responses	In *weaker responses*, candidates misclassified simple carbohydrates—contrasting them with disaccharides—as polysaccharides, oligosaccharides, or trisaccharides. Some included unrelated biomolecules such as lipids, proteins, and enzymes. Others listed various functional groups without aligning them correctly with the classification criteria.
Images of Weaker Response	Image i  Image ii  Image iii 

Suggestions for improvement (Highlight all that apply)

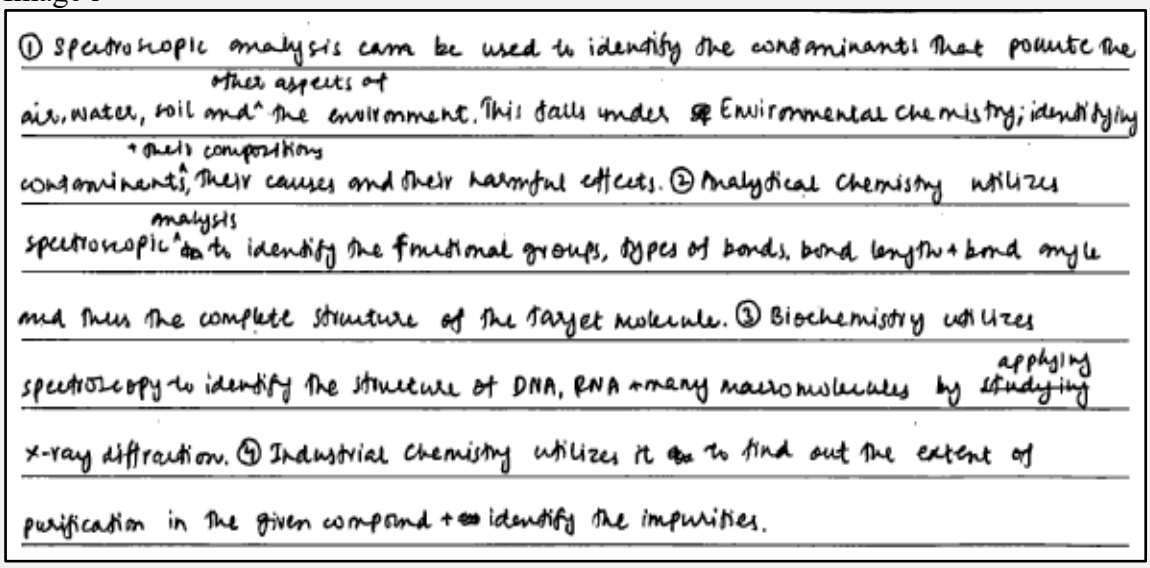
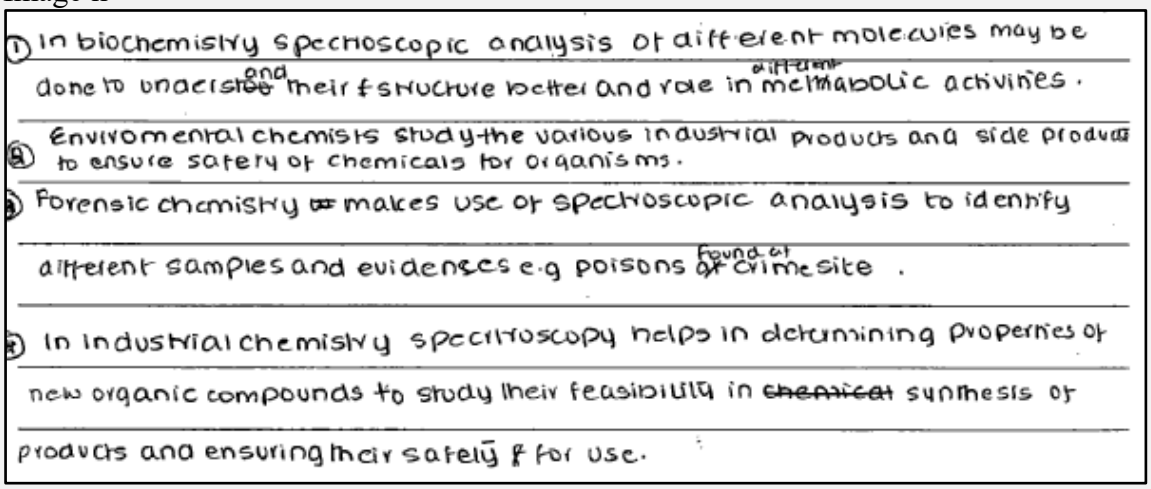
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Any Additional Suggestion:		

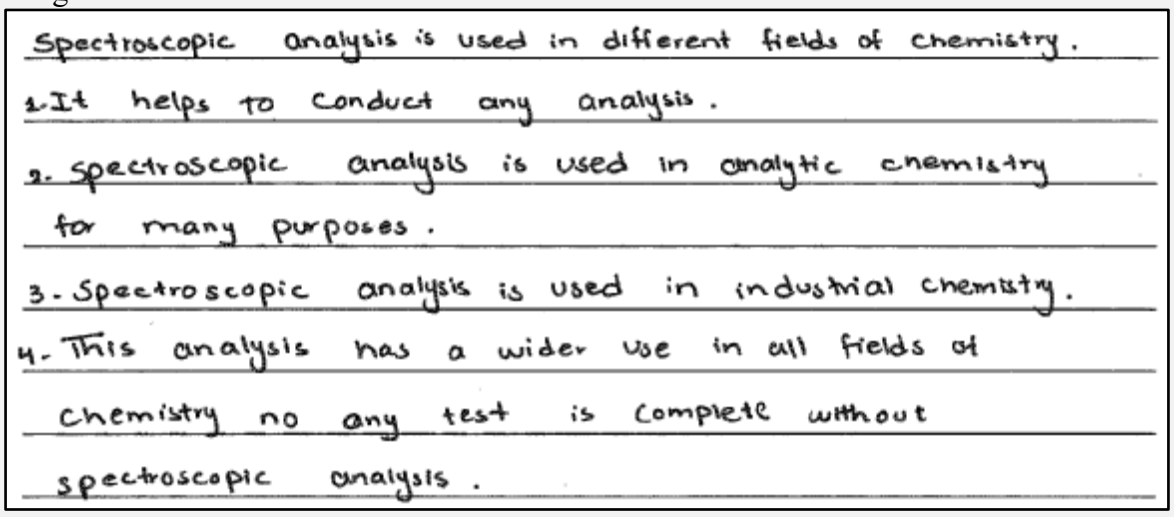
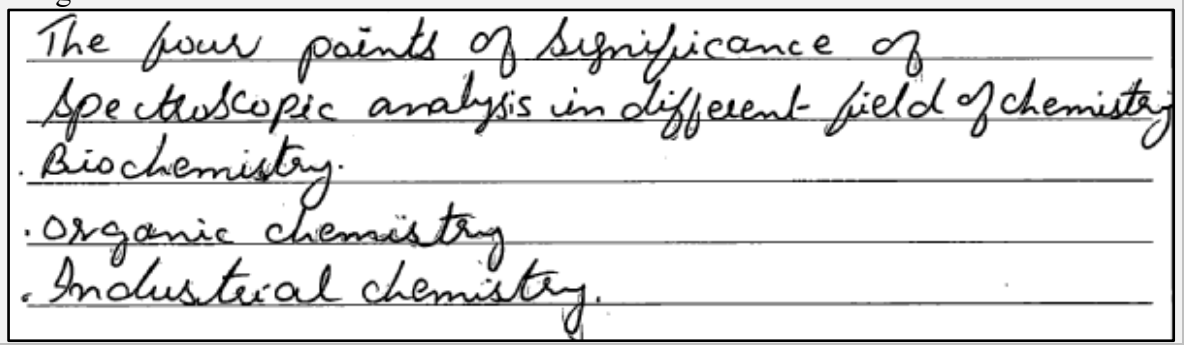
Question No. 5	
Question Text	a. Describe the process of cracking of organic compounds. b. Describe the catalytic reforming of petroleum.
SLO No.	22.4.2
SLO Text	Explain the processes of cracking (with its types) and reforming of petroleum.
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for description of cracking 2 marks for describing the catalytic reforming of petroleum 1 mark for identifying the catalyst used / any one condition 1 mark for describing the reformation of petroleum
Overall Performance	Overall, candidates' performance in this question remained moderate. Mostly, candidates successfully described part (a) and struggled to perform in part (b).
Description of Better Responses	In part (a), *better responses* clearly described the cracking of larger organic molecules into smaller hydrocarbon chains, with some candidates illustrating the process using chemical equations. In part (b), candidates effectively explained catalytic reforming by outlining the process, naming the catalyst, and stating its purpose—improving the octane rating of petroleum. Many also included relevant pressure and temperature conditions.
Images of Better Responses	Part 'a' Image i  Cracking is the process in which large hydrocarbon ~~compound~~ chains are broken down into smaller hydrocarbon chains to either (i) increase the octane no. of petrol increase or (ii) yield gasoline petrol + alkenes Image ii  cracking or pyrolysis is breaking of long chain hydrocarbons into small hydrocarbons in absence of oxygen. ex- $C_{10}H_{22} \longrightarrow C_8H_{18} + C_2H_4$ Part 'b' Image i,  Catalytic Reforming occurs when straight chain hydrocarbons are converted into cycloalkanes or branched or aromatic hydrocarbons in order to increase the octane no. of petrol and yield high quality petrol. This would occur at low temperatures in the presence of catalysts like Rhodium or Platinum. Image ii  Reforming or aromatisation involves conversion of open chain compounds into aromatic ones- catalytic reforming occurs at low Temp and low pressure 635°C, 2atm in the Presence of catalyst - mostly zeolite catalyst is used - catalytic reforming yields gasoline with high octane number-
Description of Weaker Responses	In *weaker responses*, in part (a), candidates provided vague or incomplete explanations of cracking, often omitting the concept of breaking larger hydrocarbons into smaller ones. Candidates frequently confuse cracking with polymerisation. In part (b), many failed to describe catalytic reforming accurately, with some confusing it with cracking or other

	refining processes. Key elements such as the catalyst, purpose, and operating conditions were frequently missing or incorrectly stated. Many described unrelated processes, such as fractional distillation, destructive distillation of coal, or the formation of crude oil, rather than focusing on catalytic reforming of petroleum.
Images of Weaker Responses	Part 'a' Image i  Image ii  Part 'b' Image i  Image ii 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:		

Question No. 6	
Question Text	Write any FOUR points of significance of spectroscopic analysis in different fields of chemistry.
SLO No.	24.1.4
SLO Text	Discuss application of spectroscopy in analytical chemistry.
Max Marks	4
Cognitive Level	U
Checking Hints	1 mark for each significance (any 4 required)
Overall Performance	The overall performance on this question was satisfactory. Notably, some candidates demonstrated a true understanding of various spectroscopic techniques and their respective significance in different fields of chemistry.
Description of Better Responses	*Better responses* demonstrated a comprehensive understanding of the application of spectroscopic techniques in structural analysis, concentration and purity determination, detection of elemental and heavy metals, compound colour analysis, medical imaging and screening, reaction monitoring, mechanism elucidation, and pollutant identification. Candidates effectively related these uses to various scientific disciplines, particularly chemistry, such as food science, industrial chemistry, pharmaceuticals, environmental impact assessment, forensic and medical sciences, biochemical and analytical sciences, and research and development. Many also connected their significance to different branches of chemistry, which was considered appropriate.
Images of Better Response	Image i  ① Spectroscopic analysis can be used to identify the contaminants that pollute the air, water, soil and other aspects of the environment. This falls under Environmental chemistry; identifying contaminants, their compositions, their causes and their harmful effects. ② Analytical chemistry utilizes spectroscopic analysis to identify the functional groups, types of bonds, bond length + bond angle and thus the complete structure of the target molecule. ③ Biochemistry utilizes spectroscopy to identify the structure of DNA, RNA + many macromolecules by applying x-ray diffraction. ④ Industrial chemistry utilizes it to find out the extent of purification in the given compound + identify the impurities. Image ii  ① In biochemistry spectroscopic analysis of different molecules may be done to understand their structure better and role in different metabolic activities. ② Environmental chemists study the various industrial products and side products to ensure safety of chemicals for organisms. ③ Forensic chemistry makes use of spectroscopic analysis to identify different samples and evidences e.g poisons found at crime site. ④ In industrial chemistry spectroscopy helps in determining properties of new organic compounds to study their feasibility in ~~chemical~~ synthesis of products and ensuring their safety for use.

Description of Weaker Responses	*Weaker responses* reflected a limited understanding of the significance of spectroscopy. Candidates primarily focused on comparing spectroscopy with classical methods of structural analysis, rather than explaining the respective applications. Additionally, many merely listed branches of chemistry without detailing the role or importance of spectroscopic techniques within those fields.
Images of Weaker Response	Image i  Spectroscopic analysis is used in different fields of chemistry. 1. It helps to conduct any analysis. 2. spectroscopic analysis is used in analytic chemistry for many purposes. 3. Spectroscopic analysis is used in industrial chemistry. 4. This analysis has a wider use in all fields of chemistry no any test is complete without spectroscopic analysis. Image ii  The four points of significance of spectroscopic analysis in different field of chemistry . Biochemistry. . Organic chemistry . Industrial chemistry.

Suggestions for improvement (Highlight all that apply)

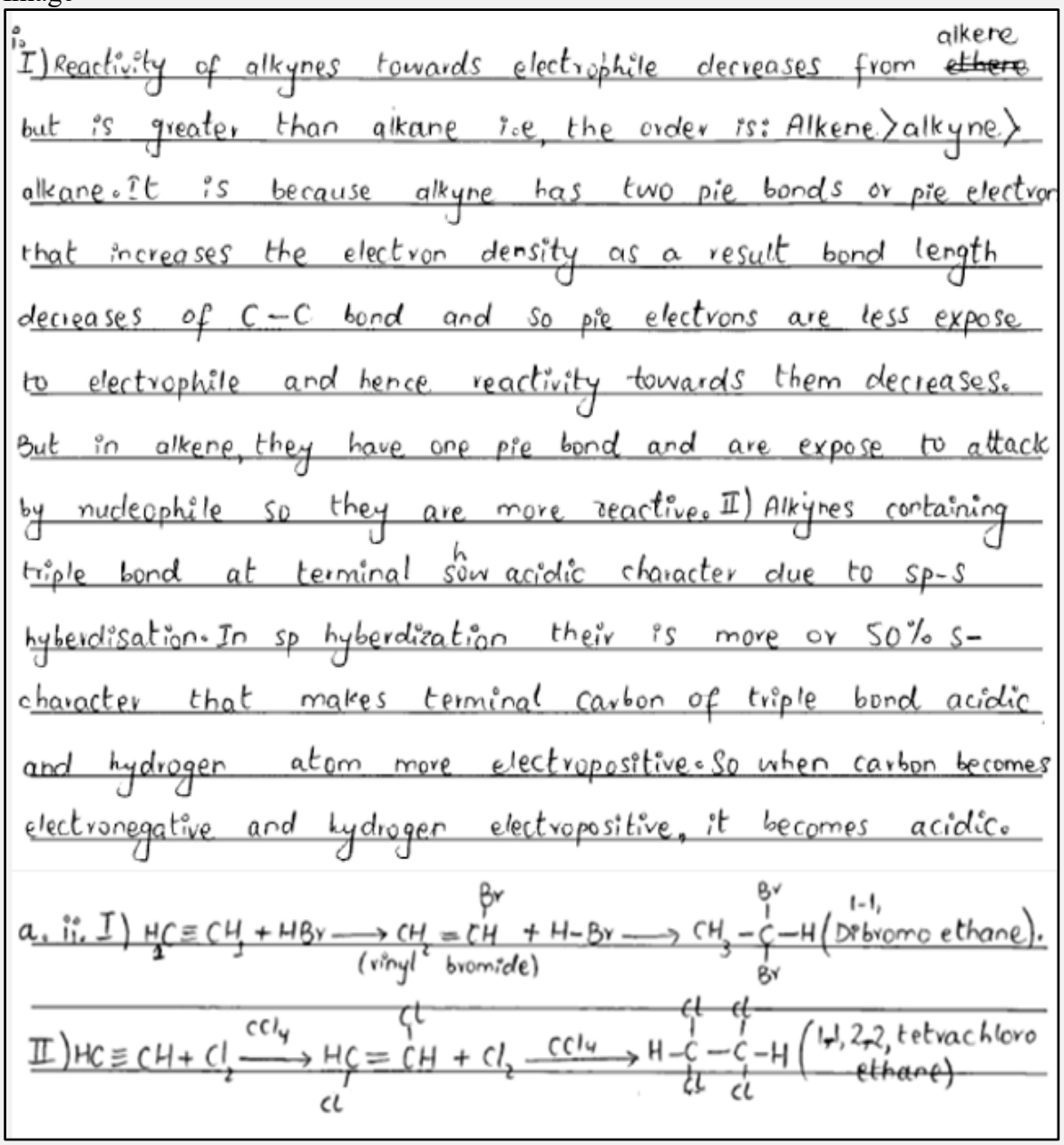
Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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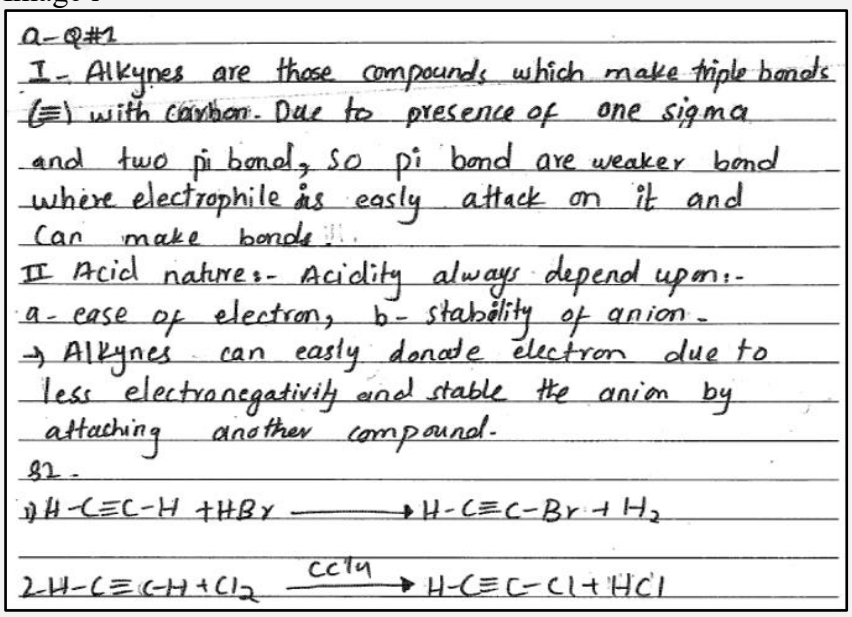
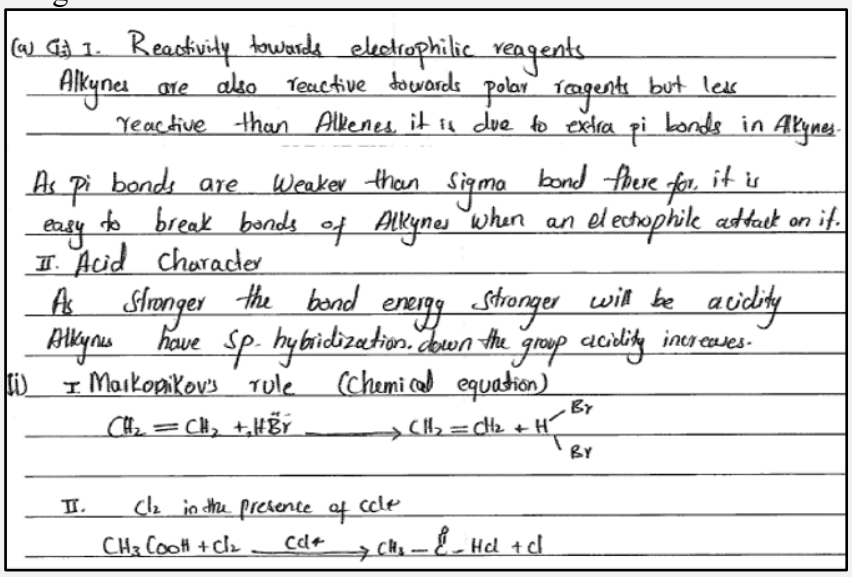
Any Additional Suggestion: Teachers should enable students to differentiate between fields and branches of chemistry. This can be done by relating various topics, SLOs, which are based on uses and applications and examples linking spectroscopic aspects with them.

Extended Response Questions (ERQs)

Extended response questions offered a choice between parts 'a' and 'b'

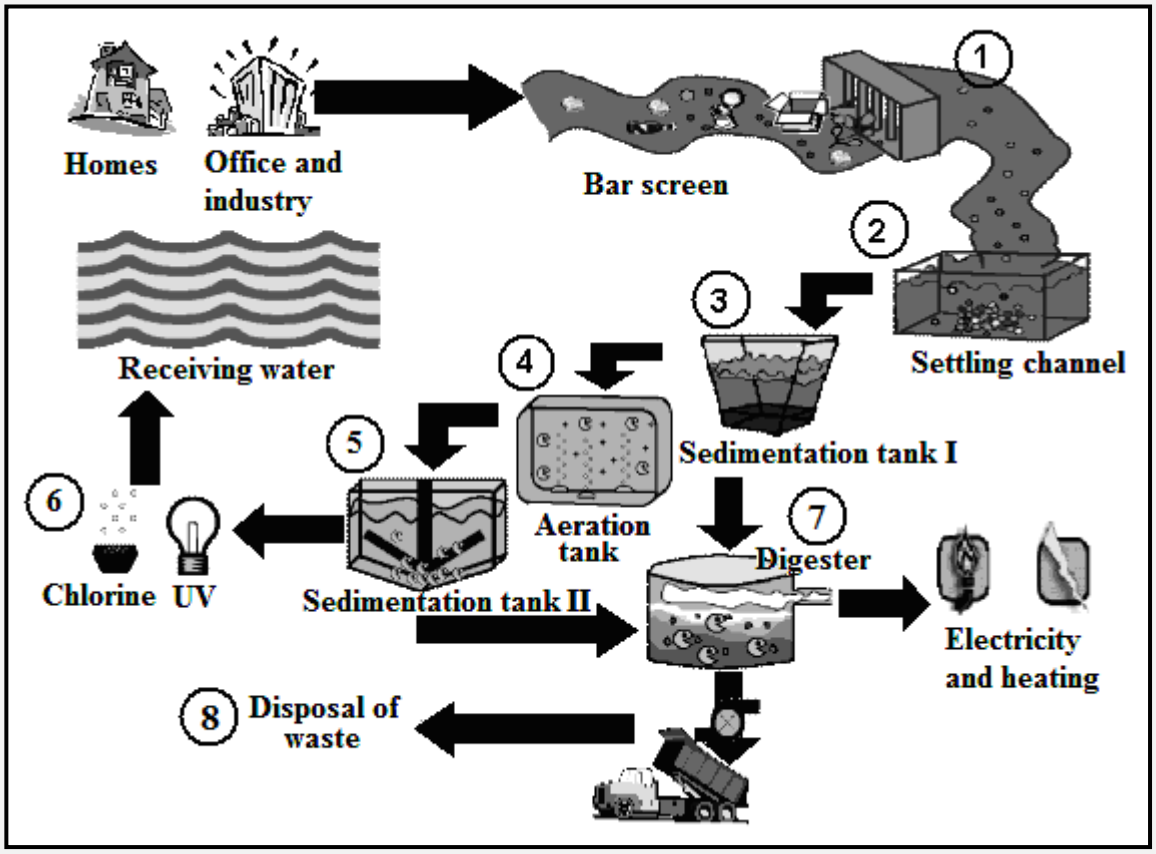
Question No. 7a	
Question Text	i. Describe the following characteristics of alkynes. I. Reactivity towards electrophilic reagents II. Acidic character ii. Using balanced chemical equations, show complete reactions of ethyne (C_2H_2) with the following reactants. I. Hydrogen bromide (HBr) in accordance with Markovnikov's rule II. Chlorine gas (Cl_2) in the presence of carbon tetrachloride (CCl_4)
SLO No.	16.5.1, 16.4.2
SLO Text	Compare the reactivities of alkynes, alkenes and alkanes. Describe the reactions of ethene, i.e., a. hydrogenation b. hydration c. hydrohalogenation d. halogenation e. halohydration f. epoxidation g. ozonolysis h. polymerisation.
Max Marks	7
Cognitive Level	U
Checking Hints	i. I. 1 mark for writing about bonding in alkyne 1 mark for writing about electron density/ electrons are less exposed/ less reactive toward electrophilic reagents than alkene/ more than alkane II. 1 mark for writing about sp-s overlap/ s-character 1 mark for writing about the pulling of electrons, making acidic hydrogen ii. 2 marks for writing a two-step equation with HBr *Candidate may write the first equation in a single step with the correct final product.* *i.e., geminal dibromide; 2 Br atoms on one carbon)* 1 mark for writing the equation with chlorine
Overall Performance	Comparatively fewer candidates opted for this part of ERQ, but generally performed well on this question and elaborated all the steps of the chemical equation and justified the reasoning.
Description of Better Responses	*Better responses*, in section (I) of part (i), demonstrated a strong conceptual understanding of the chemical behaviour of alkynes. Candidates accurately explained that in alkynes, carbon atoms are connected by a triple bond consisting of one sigma and two pi bonds. They recognised that the high electron density within this bond pulls the atoms closer together, making the electrons less exposed. As a result, alkynes are less reactive towards electrophilic reagents than alkenes, but more reactive than alkanes. In addition, many candidates, in section (II) of part (i), correctly described the nature of the terminal hydrogen in alkynes. They noted that the hydrogen is attached via an sp–s overlap, and that the sp hybrid orbital, with its 50% s character, renders the carbon more electronegative than in sp^2 or sp^3 hybridisation. This increased electronegativity draws electron density more strongly, making the terminal hydrogen slightly acidic. For part (ii), stronger responses included correct chemical equations: • A two-step or one-step reaction forming a geminal dihalide.

	• A one-step reaction forming a vicinal dihalide. These candidates accurately applied mechanistic and structural understanding to the chemical transformations of alkynes.
Image of Better Response	Image  i. I) Reactivity of alkynes towards electrophile decreases from ~~ethere~~ alkene but is greater than alkane i.e, the order is: Alkene > alkyne > alkane. It is because alkyne has two pie bonds or pie electron that increases the electron density as a result bond length decreases of C–C bond and so pie electrons are less expose to electrophile and hence reactivity towards them decreases. But in alkene, they have one pie bond and are expose to attack by nucleophile so they are more reactive. II) Alkynes containing triple bond at terminal show acidic character due to sp-s hyberdisation. In sp hyberdization their is more or 50% s-character that makes terminal carbon of triple bond acidic and hydrogen atom more electropositive. So when carbon becomes electronegative and hydrogen electropositive, it becomes acidic. a. ii. I) $HC\equiv CH + HBr \longrightarrow \underset{\text{(vinyl bromide)}}{CH_2=CH(Br)} + H-Br \longrightarrow CH_3-\underset{Br}{\overset{Br}{C}}-H$ (1-1, Dibromo ethane). II) $HC\equiv CH + Cl_2 \xrightarrow{CCl_4} \underset{Cl}{HC=CH(Cl)} + Cl_2 \xrightarrow{CCl_4} H-\underset{Cl}{\overset{Cl}{C}}-\underset{Cl}{\overset{Cl}{C}}-H$ (1,1,2,2, tetrachloro ethane)
Description of Weaker Responses	*Weaker responses,* in section I of part (i), reflected a limited understanding of the structural and chemical behaviour of alkynes. Many candidates failed to explain the nature of the triple bond or its composition of one sigma and two pi bonds. The impact of high electron density, drawing carbon atoms closer and reducing reactivity towards electrophilic reagents, was often omitted or inaccurately described. Additionally, in section II of part (i), most candidates were unable to explain the acidic character of the terminal hydrogen in alkynes. The concept of sp hybridisation, its 50% s character, and its influence on electronegativity and acidity were either missing or poorly understood. In part (ii), a number of responses lacked correct chemical equations of hydrohalogenation and halogenation in alkynes. The distinction between geminal and vicinal dihalides was frequently unclear, and many equations did not reflect the required steps or product orientation, for example, 1,2-dibromoethane, because of a single-step reaction, indicating gaps in reaction knowledge, which killed the purpose of asking Markovnikov's rule.

Images of Weaker Response	Image i  Image ii 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:		

Question No. 7b	
Question Text	The figure below shows the stages involved in sewage treatment. Using the figure below, describe in detail the complete process through which sewage water is cleaned. 
SLO No.	23.2.2
SLO Text	Explain the methods of water purification, i.e. raw water treatment, sewage treatment, zeolite process and reverse osmosis;
Max Marks	7
Cognitive Level	U
Checking Hints	Give ONE mark for writing about the removal of large rubbish/grit, and sand. Give ONE mark for writing about oxygen supply in the aeration tank/ removal of foul smell Give ONE mark for writing about oxidation by bacteria. Give ONE mark for writing about the settling of sludge in sedimentation tanks I and II. Give ONE mark for writing about disinfection by chlorine / UV rays. Give ONE mark for writing about the production of methane gas/the generation of electricity/ aerobic and anaerobic digestion by bacteria Give ONE mark for writing about the burning/dumping/ selling of digested sludge.
Overall Performance	The majority of the population attempted this part of the ERQ. Candidates drafted responses satisfactorily. However, a few candidates described a few steps in the process shown in the diagram.
Description of Better Responses	*Better responses* demonstrated a clear and sequential understanding of the sewage treatment process. Candidates explained that the removal of large debris, which ensures smoother flow in the subsequent stages, is followed by the settling of grit and sand as the water flows slowly, allowing heavier particles to settle out. The process then continues with the gradual sinking of smaller waste particles, forming unpleasant odour-producing sludges. This sludge-rich water then moves into the next phase, where special bacteria play a vital role in oxidising harmful organic compounds, with a continuous supply of air to maintain aerobic

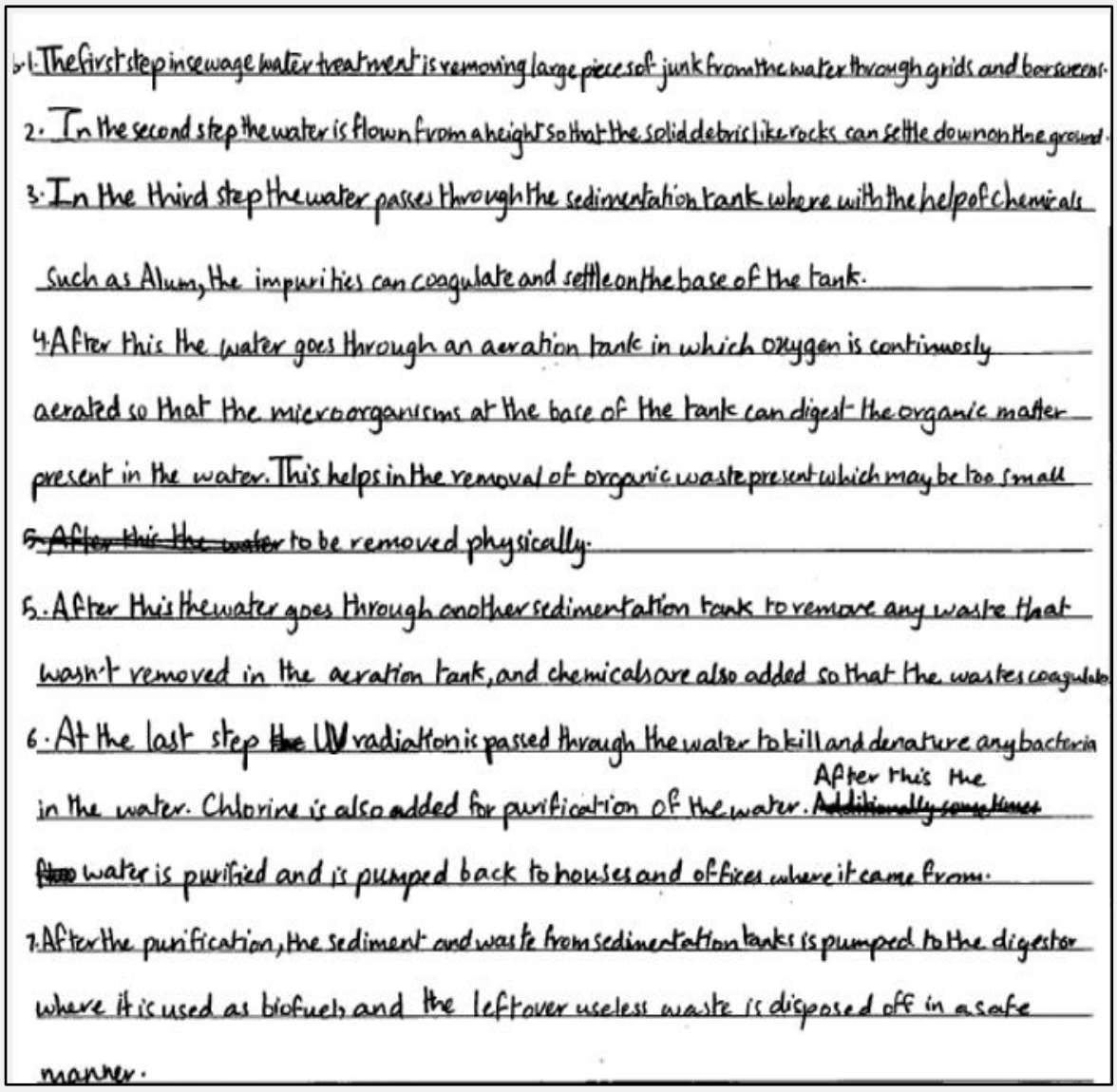
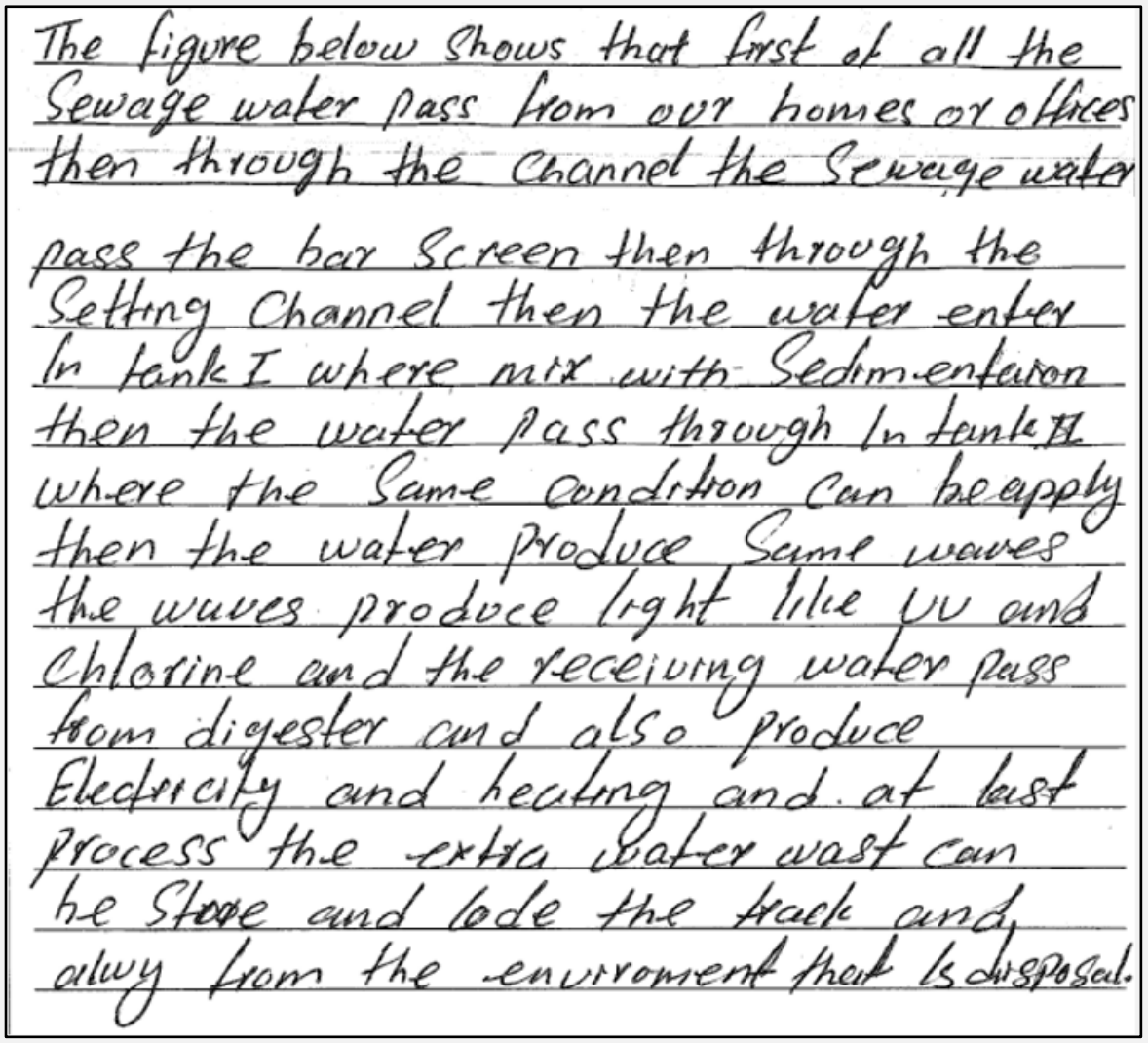
	conditions. As the treated water progresses, any remaining sludge settles further, resulting in water that is safe to discharge. Candidates correctly noted the use of disinfectants like chlorine or UV radiation to eliminate any residual harmful microorganisms. They also mentioned that the collected sludge is transferred to the next stage, where bacterial action, either aerobic or anaerobic, breaks down harmful substances and produces methane gas, which can be used to generate electricity for the plant. The final step involves waste disposal through burning, landfilling, or repurposing as fertiliser for agricultural use.
Image of Better Response	
Description of Weaker Responses	*Weaker responses* typically lacked completion and clarity in describing the full sewage treatment process. While many candidates addressed the initial stages correctly, they often failed to explain the processes occurring in the later stages. The function of sedimentation tank II, where remaining sludge settles and water becomes safe for discharge, was frequently omitted. Similarly, the role of the aeration stage, particularly the action of bacteria in oxidising harmful organic compounds and the importance of continuous air supply, was either vaguely described or entirely missing. The digester stage, involving further bacterial breakdown of sludge and methane production, was also commonly overlooked. In addition, the final step of waste disposal through incineration, landfilling, or use as fertiliser was rarely mentioned, indicating a limited understanding of the complete treatment cycle.

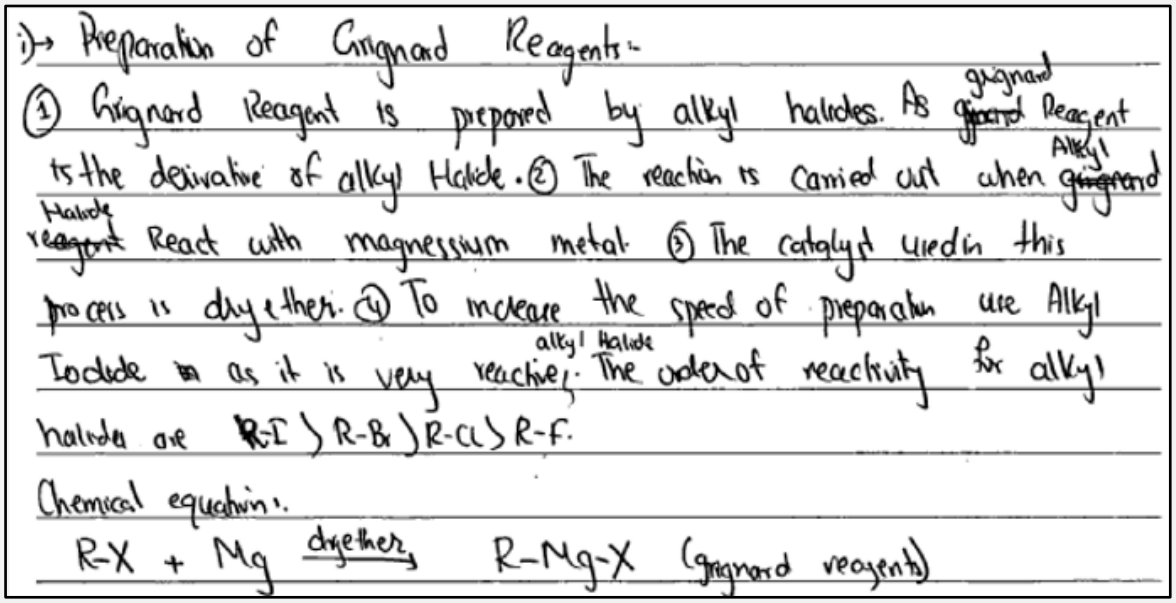
Image of Weaker Response	

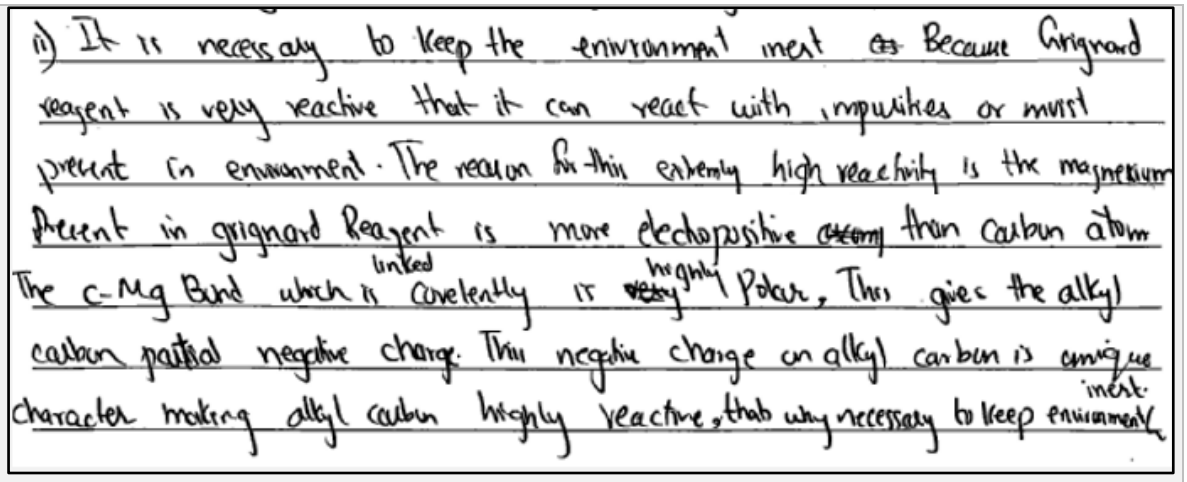
The figure below shows that first of all the Sewage water pass from our homes or offices then through the channel the Sewage water pass the bar Screen then through the Setting Channel then the water enter In tank I where mix with Sedimentation then the water pass through In tank II where the same condition can be apply then the water produce Same waves the waves produce light like UV and Chlorine and the receiving water pass from digester and also produce Electricity and heating and at last process the extra water wast can be Store and lode the track and alwy from the environment that is disposal.

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion: The schematic illustration of the treatment plants using charts and post-its can be used for practising such long and sequential processes.

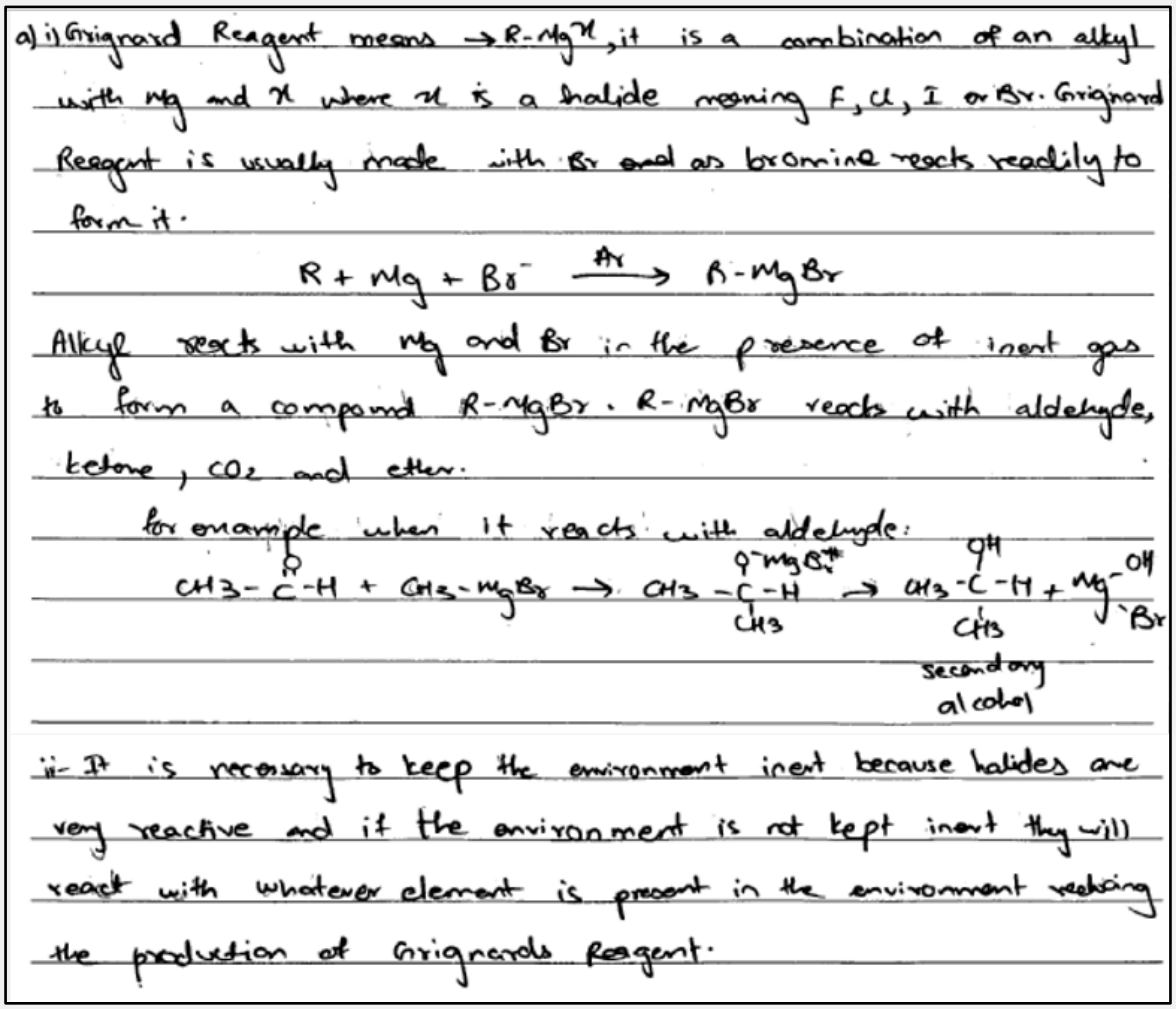
Question No. 8a	
Question Text	i. Describe the preparation of Grignard reagents in FOUR points. Support your answer with a chemical equation. ii. Give a reason why it is necessary to keep the environment inert during the synthesis of Grignard's reagent.
SLO No.	17.4.1
SLO Text	Describe the preparation and reactivity of Grignard reagent.
Max Marks	7
Cognitive Level	U
Checking Hints	i. 1 mark for describing each point of preparation (any 4 points required) 1 mark for describing the use of magnesium with organic halides 1 mark for describing the use of ether/ dry and inert conditions 1 mark for describing the exothermic nature of the reaction 1 mark for describing the use of a boiling solvent/ electric heater to initiate the reaction 1 mark for describing the reactivity of organic halides 1 mark for writing a chemical equation to show the preparation of the Grignard reagent *Any other relevant example will be awarded 1 mark* ii. 1 marks for writing the reason for the inert environment (any TWO required)
Overall Performance	Most candidates opted for this part of the ERQ. Overall, candidates showed unsatisfactory performance in this question. Most candidates were unable to gauge the requirements of the question. Instead, they explained the reactions of Grignard's reagent.
Description of Better Responses	*Better responses* demonstrated a clear understanding of the preparation of Grignard's reagents by reacting magnesium with alkyl or aryl halides in dry ether. Candidates correctly highlighted ether's role in stabilising the organo-magnesium compound and described the use of appropriate apparatus to prevent exposure to air and moisture. Most of the candidates also mention the reactivity of halides that are suitable for use in the preparation, i.e., Cl and Br. They acknowledged the slow induction followed by an exothermic reaction and noted bromides as the preferred halides due to moderate reactivity, with dry ether used for less reactive aryl halides. Candidates also mentioned a chemical equation using various alkyl halides. Many included the correct chemical equation and stressed the importance of air- and moisture-free conditions, explaining that exposure leads to oxidation or hydrolysis, destroying the reagent. while reaction with water results in the formation of alkanes and magnesium hydroxide, rendering the Grignard reagent ineffective.
Image of Better Response	


ii) It is necessary to keep the environment inert ~~as~~ Because Grignard reagent is very reactive that it can react with impurities or moist present in environment. The reason for this extremly high reactivity is the magnesium present in grignard Reagent is more electropositive ~~atom~~ than carbon atom linked The C-Mg Bond which is covelently is ~~very~~ highly Polar, This gives the alkyl carbon partial negative charge. This negative charge on alkyl carbon is anionic inert. character making alkyl carbon highly reactive, that why necessary to keep environment

Description of Weaker Responses

Weaker responses commonly lacked clarity regarding the preparation of Grignard's reagents. Many candidates incorrectly focused on the reactions involving Grignard reagents—such as with aldehydes, ketones, and carboxylic acids—which relate to a different SLO and were not relevant to the question. A significant number were unable to present the correct chemical equation for the preparation, with some incorrectly showing an alkane reacting with magnesium halide. Additionally, candidates failed to explain the necessity of maintaining an inert reaction environment, often attributing it incorrectly to the reactivity of magnesium rather than the actual sensitivity of the Grignard reagent to air and moisture.

Image of Weaker Response


a) i) Grignard Reagent means → R-MgX, it is a combination of an alkyl with Mg and X where X is a halide meaning F, Cl, I or Br. Grignard Reagent is usually made with Br and as bromine reacts readily to form it.

$$R + Mg + Br^- \xrightarrow{Ar} R\text{-}MgBr$$

Alkyl reacts with Mg and Br in the presence of inert gas to form a compound R-MgBr. R-MgBr reacts with aldehyde, ketone, CO_2 and ether.

for example when it reacts with aldehyde:

$$CH_3\text{-}\overset{O}{\overset{\|}{C}}\text{-}H + CH_3\text{-}MgBr \rightarrow CH_3\text{-}\overset{O^-MgBr^+}{\underset{CH_3}{C}}\text{-}H \rightarrow CH_3\text{-}\underset{CH_3}{\overset{OH}{C}}\text{-}H + Mg^{-OH}_{Br}$$

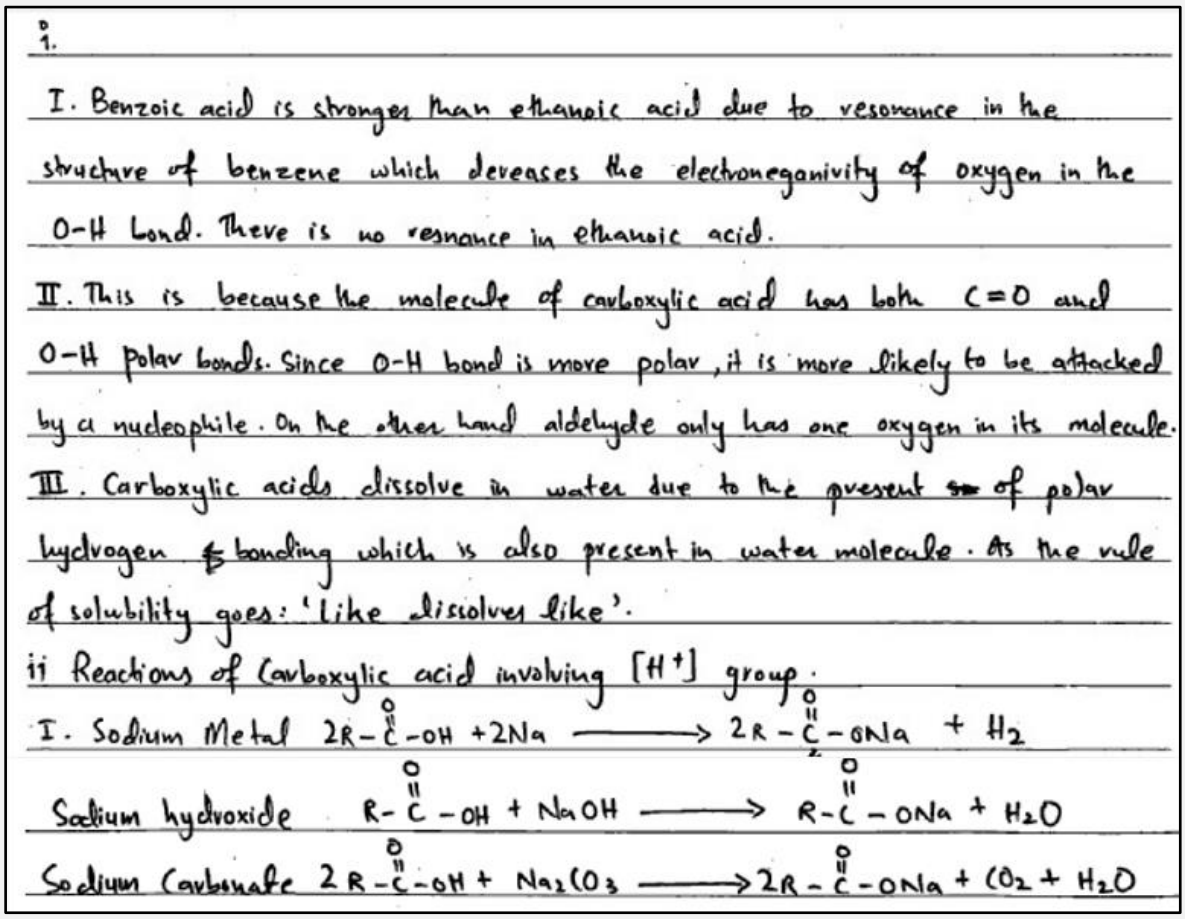
secondary alcohol

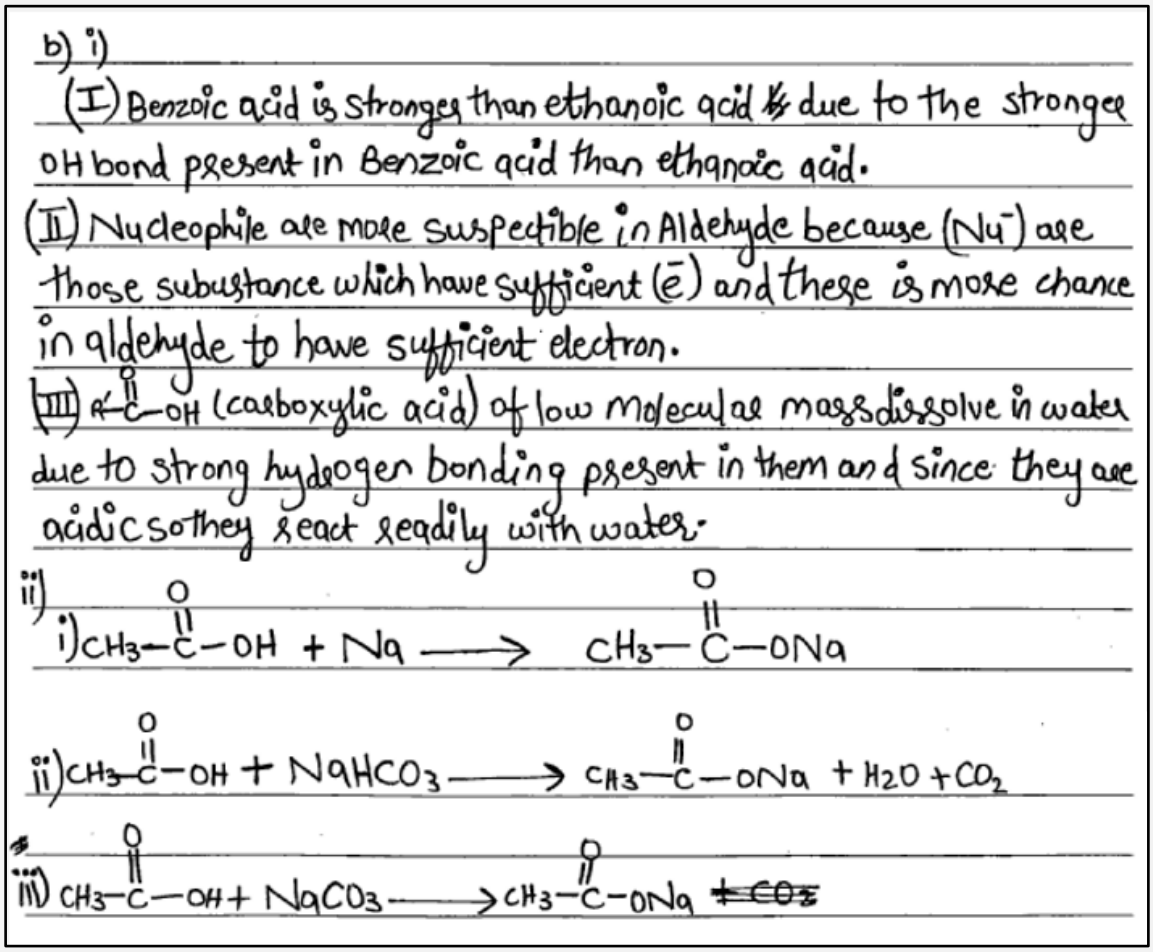
ii- It is necessary to keep the environment inert because halides are very reactive and if the environment is not kept inert they will react with whatever element is present in the environment reducing the production of Grignards Reagent.

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:		

Question No. 8b	
Question Text	i. Give a reason for each of the following statements. I. Benzoic acid is stronger than ethanoic acid. II. The carbonyl group in a carboxylic acid is less susceptible to nucleophile than aldehyde. III. Carboxylic acids of fairly low molecular masses dissolve in water. ii. Write the general chemical equations to show the reactions of carboxylic acid with the given substances. I. Sodium metal II. Sodium hydroxide III. Sodium carbonate
SLO No.	20.3.1 20.5.1 20.2.1
SLO Text	Discuss the acidity of carboxylic acids. Describe the reactions of carboxylic acid involving a. hydrogen atom of the carboxyl group b. hydroxyl group of carboxyl group c. carboxyl group as a whole. Describe the structure and physical properties (solubility, melting point and boiling point) of carboxylic acid.
Max Marks	7
Cognitive Level	U*
Checking Hints	i. 1 mark for the correct reason 1 mark for writing each point (TWO required) 1 mark for the correct reason ii. 1 mark for writing each equation (3 required)

Overall Performance	An equal number of candidates who attempted this part of ERQ showed satisfactory performance. The responses depicted a lack of an in-depth understanding of the comparative reasoning of carboxylic acid and its reaction.
Description of Better Responses	*Better responses*, in the first part, demonstrated a comprehensive understanding of the chemical properties and reactivity of carboxylic acids. Candidates accurately explained that the greater delocalisation of the negative charge in the benzoate ion ($C_6H_5COO^-$) across the benzene ring enhances the stability of the molecule due to resonance. They supported their explanations with correct representations of resonating structures. Candidates also recognised that in carboxylic acids, the carbon is sp^2-hybridised, similar to that in aldehydes. However, the lone pair delocalisation from the hydroxyl oxygen into the carbonyl group reduces the partial positive charge (δ^+) on the carbon atom, making the C=O bond less electrophilic and less susceptible to nucleophilic attack. Additionally, they explained that the electron-withdrawing effect of the carbonyl group increases the polarity of the O–H bond, making the hydrogen more easily ionisable as H^+, hence contributing to the acidic character of carboxylic acids. Regarding solubility, better responses correctly stated that lower molecular mass carboxylic acids are more polar and form hydrogen bonds with water, making them more soluble. Candidates noted that smaller carboxylic acid molecules face less steric hindrance and weaker van der Waals interactions, enhancing solubility in water. In the second part, candidates appropriately identified the acidic behaviour of carboxylic acids through their reactions with metals and alkalis. They correctly wrote balanced equations showing the formation of carboxylate ions when reacting with sodium metal, sodium hydroxide, and sodium carbonate
Image of Better Responses	 1. I. Benzoic acid is stronger than ethanoic acid due to resonance in the structure of benzene which deveases the electroneganivity of oxygen in the O-H bond. There is no resonance in ethanoic acid. II. This is because the molecule of carboxylic acid has both C=O and O-H polar bonds. Since O-H bond is more polar, it is more likely to be attacked by a nucleophile. On the other hand aldehyde only has one oxygen in its molecule. III. Carboxylic acids dissolve in water due to the present of polar hydrogen bonding which is also present in water molecule. As the rule of solubility goes: 'Like dissolves like'. ii Reactions of Carboxylic acid involving [H+] group. I. Sodium Metal $2R-\overset{O}{\overset{\|}{C}}-OH + 2Na \longrightarrow 2R-\overset{O}{\overset{\|}{C}}-ONa + H_2$ Sodium hydroxide $R-\overset{O}{\overset{\|}{C}}-OH + NaOH \longrightarrow R-\overset{O}{\overset{\|}{C}}-ONa + H_2O$ Sodium Carbonate $2R-\overset{O}{\overset{\|}{C}}-OH + Na_2CO_3 \longrightarrow 2R-\overset{O}{\overset{\|}{C}}-ONa + CO_2 + H_2O$
Description of Weaker Responses	*Weaker responses* generally lacked coherence in addressing both reasoning and equation-based components of the question. Many candidates were unable to justify the acidic behaviour or solubility of carboxylic acids effectively. Several responses also showed confusion between aldehydes and carboxylic acids, particularly in describing electrophilicity

	and nucleophilic reactivity, further indicating conceptual gaps. Some candidates attempted the reasoning but failed to support their explanations with correct chemical equations. Others, conversely, wrote relevant equations, such as reactions with sodium metal, sodium hydroxide, or sodium carbonate, but did not link them to the underlying concepts of resonance, polarisation, or molecular interactions.
Image of Weaker Response	

b) i)

(I) Benzoic acid is stronger than ethanoic acid due to the stronger OH bond present in Benzoic acid than ethanoic acid.

(II) Nucleophile are more suspectible in Aldehyde because (Nu^-) are those subustance which have sufficient ($\bar{e}$) and these is more chance in aldehyde to have sufficient electron.

(III) R–C(=O)–OH (carboxylic acid) of low molecular mass dissolve in water due to strong hydrogen bonding present in them and since they are acidic so they react readily with water.

ii)

i) $CH_3-C(=O)-OH + Na \longrightarrow CH_3-C(=O)-ONa$

ii) $CH_3-C(=O)-OH + NaHCO_3 \longrightarrow CH_3-C(=O)-ONa + H_2O + CO_2$

iii) $CH_3-C(=O)-OH + NaCO_3 \longrightarrow CH_3-C(=O)-ONa$ ~~$+ CO_3$~~

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login
Any Additional Suggestion:		

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

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